

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B.Tech (Mechanical) (CBCS) Degree Examinations

December 2023

ME1104-1 – THERMAL ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

- Use of thermodynamic data hand book is permitted.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- An adiabatic system is**
 - An open system insulated at its real boundaries
 - A system in which heat energy interaction takes place between system and surroundings
 - An closed system insulated at its real boundaries
 - An isolated system
- Internal combustions are an example of**
 - Closed system
 - Open system
 - Adiabatic system
 - Isolated system
- The path functions are regarded as**
 - Inexact differentials
 - Exact differentials
 - Perfect differentials
 - Path independent
- For a reversible process 1-2 in which pressure=constant, work done is given by**
 - Zero
 - $p_1(V_2 - V_1)$
 - $mRT_1 \ln(V_2 / V_1)$
 - $(P_1V_1 - P_2V_2)/(n-1)$
- The relationship between heat and work form of energies for a closed system undergoing a reversible process is**
 - $W=Q+\Delta E$
 - $W=\Delta U+Q$
 - $Q=\Delta E+W$
 - $Q=\Delta U+W$
- What is the role of a nozzle ?**
 - Increases the pressure of fluid at the expense of its kinetic energy
 - Increases the kinetic energy of fluid at the expense of its pressure
 - Decreases the pressure and kinetic energy of fluid
 - Increases the pressure of fluid and also its kinetic energy
- The thermal energy reservoir which absorbs heat continuously without suffering changes in its temperature is called as**
 - High temperature reservoir
 - Sink
 - Source
 - Heat engine
- The COP of Carnot heat pump is**
 - $\frac{T_{max}}{T_{max} - T_{min}}$
 - $\frac{T_{min}}{T_{max} - T_{min}}$
 - $1 - \frac{T_{min}}{T_{max}}$
 - $\frac{T_{max}}{T_{min} - T_{max}}$
- Rankine cycle comprises of**
 - two isentropic processes and two constant volume processes.
 - two isothermal processes and two constant pressure processes.
 - two isentropic processes and two constant pressure processes.
 - two isothermal processes and two constant volume processes.

10. Calculate the thermal efficiency of vapour power plant if the net work done is 500 kJ/kg and heat addition is 2000 kJ/kg.
A) 2.5 %
B) 66.67%
C) 25 %
D) 250 %
11. Steam rate of a vapour power cycle is defined as
A) Rate of steam flow in kg/hour required to produce unit power output of 1 kW.
B) Rate of steam flow in kg/s required to produce unit shaft output of 1 kW.
C) Rate of steam flow in kg/hour required to produce unit shaft output of 1 kJ/hour.
D) Rate of steam flow in kg/min required to produce unit shaft output of 1 kW.
12. The air standard efficiency of Otto cycle (η_{OTTO}) is given by
A) $1 - \frac{1}{r^\gamma}$
B) $1 - \frac{1}{r^{\gamma-1}}$
C) $1 - \frac{1}{r^{\gamma+1}}$
D) $1 - \frac{1}{r^{\frac{(\gamma-1)}{\gamma}}}$
13. The mean effective pressure of an air standard cycle is given by
A) $\frac{\text{Net work done per cycle}}{\text{Total volume of the cylinder}}$
B) $\frac{\text{Net work done per cycle}}{\text{Swept volume of the cylinder}}$
C) $\frac{\text{Net work done per cycle}}{\text{Clearance volume of the cylinder}}$
D) $\frac{\text{Heat added per cycle}}{\text{Swept volume of the cylinder}}$
14. Calculate the thermal efficiency of Brayton cycle with pressure ratio of 8.
A) 56.47 %
B) 47.5 %
C) 44.8 %
D) 50.16 %
15. The compression ratio of a spark ignition engine is in the range of
A) 4 to 6
B) 6 to 8
C) 6 to 10
D) 10 to 14
16. The Dual cycle consists of
A) Two reversible isothermal processes and two reversible isobaric processes
B) Two reversible isochoric process and two reversible adiabatic processes and one reversible isobaric process.
C) One reversible isothermal process and two reversible isochoric process and one reversible isobaric process.
D) Two reversible isobaric processes and two reversible adiabatic processes
17. A humidification process means
A) a decrease in relative humidity
B) a decrease in specific humidity
C) an increase in specific humidity
D) an increase in temperature
18. The COP of a refrigerator is given by
A) $\frac{(\text{Cooling effect})}{(\text{Work input})}$
B) $\frac{(\text{Work input})}{(\text{Cooling effect})}$
C) $\frac{(\text{Heat added})}{(\text{Cooling effect})}$
D) $\frac{(\text{Cooling effect})}{(\text{Heat added})}$
19. Identify the irreversible process in a vapour compression refrigeration unit.
A) Isentropic compression.
B) Constant pressure heat absorption.
C) Throttling.
D) Constant pressure heat rejection.
20. The relation for relative humidity (ϕ) of air is given as
A) $\frac{p_s}{p_w}$
B) $\frac{p_s}{p}$
C) $\frac{p}{p_s}$
D) $\frac{p_w}{p_s}$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Distinguish between:				
	i) Intensive and extensive properties.	4	L*1	1	1
	ii) Open system and closed system.	4	L1	1	1
b)	State zeroth law of thermodynamics. Write its importance in thermodynamics.				
c)	Air flows steadily at the rate of 0.5 kg/s through an air compressor, entering at 7 m/s velocity, 100 kPa pressure, 0.95 m ³ /kg volume, and leaving at 5 m/s velocity, 700 kPa pressure and 0.19 m ³ /kg. The internal energy of the air leaving is 93 kJ/kg greater than that of the air entering. Cooling water in the compressor jackets absorbs heat from the air at the rate of 58 kW.				
	i) Compute the rate of shaft work in kW.	8	L3	2	2
	ii) Find the ratio of the inlet pipe diameter to outlet pipe diameter.				
2. a)	What are volume flow rate and mass flow rate? Derive the relation between the two.	4	L1	1	1
b)	What is perpetual motion machine of first kind?	4	L2	1	1
c)	A reversible heat engine operates between two reservoirs at temperature of 600°C and 40°C. The engine drives a reversible refrigerator, which operates between reservoirs at temperatures of 40°C and -20 °C. The heat transfer to the heat engine is 2000 kJ and the network output of the combined engine refrigerator plant is 360 kJ. Evaluate the heat added to the refrigerant and the net heat added to the reservoir at 40°C.	8	L3	2	2
3. a)	Derive an expression for a closed system of ideal gas undergoing a polytropic process. Also obtain the relation of heat and work in the form $Q_{1-2} = \frac{(\gamma-n)}{(\gamma-1)} W_{1-2}$	8	L2	1	1
b)	Derive an expression for thermal efficiency of Carnot cycle.	8	L2	2	1
Unit – II					
4. a)	Draw a neat sketch for p-V diagram for pure substance water and indicate various regions.	5	L1	3	1
b)	Define the following with regard to pure substance water:				
	i) Saturation pressure				
	ii) Latent heat of vapourization				
	iii) Enthalpy of superheat.	3	L1	3	1
c)	A gas power plant operating on an Brayton cycle has a pressure ratio of 8. The temperature of air at compressor inlet is 27 °C and 1273 °C at the turbine inlet. Determine				
	i) Temperature of air at the exit of turbine and compressor				
	ii) Brayton efficiency of the cycle				
	iii) If the power output from the plant is 10 MW, calculate the air flow rate in kg/s.	8	L3	4	2
5. a)	A 30 MW steam plant working on Rankine cycle operates between boiler pressure of 50 bar and condenser pressure of 0.2 bar. The steam leaves the boiler and enters the steam turbine at 500°C. Determine				
	(i) Quality of steam from the turbine exit				
	(ii) Cycle efficiency and				
	(iii) Steam flow rate in kg per second.	8	L3	3	1
b)	Derive an expression for air standard efficiency of Diesel cycle with p-V and T-s property diagram.	8	L2	4	1

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|-------|--|---|----|---|---|
| 6. a) | Explain the working of Rankine cycle with the help of T-s and p-V property diagrams. | 8 | L2 | 3 | 1 |
| b) | An engine of 250 mm bore and 375 mm stroke works on constant volume cycle. The clearance volume is 0.00263 m ³ . The initial pressure and temperature are 1 bar and 50 °C. If the maximum pressure is 25 bar, find: | | | | |
| i) | Air standard efficiency. | | | | |
| ii) | The temperature and pressure at the cardinal points of the cycle. | 8 | L3 | 4 | 2 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | Explain the various processes that constitute the air refrigeration cycle with p-V and T-s property diagrams. | 6 | L2 | 5 | 1 |
| b) | Define the following with respect to psychrometry: | | | | |
| i) | Specific humidity | | | | |
| ii) | Wet bulb temperature | 2 | L1 | 5 | 1 |
| c) | A simple R-12 plant is to develop 7 tonnes of refrigeration. The condenser and evaporator temperatures are to be 30°C and -5°C respectively. Determine: | | | | |
| i) | Refrigerant flowrate in kg/s | | | | |
| ii) | Power required to drive the compressor | | | | |
| iii) | Quality of refrigerant at the inlet of evaporator | | | | |
| iv) | COP | 8 | L3 | 5 | 2 |
| 8. a) | Explain the following psychrometric processes | | | | |
| i) | Heating and humidification | | | | |
| ii) | Cooling and humidification | 8 | L2 | 5 | 1 |
| b) | Moist air at 35°C and 101.325 kPa has a dew point of 15°C. Determine the following without using psychrometric chart. | | | | |
| i) | relative humidity | | | | |
| ii) | specific humidity | | | | |
| iii) | enthalpy. | 8 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
